EXPERIMENT – 3

Title: To Perform SELECT, INSERT, UPDATE and DELETE Operations on a Table

Aim / Objective

To understand and perform the basic **Data Manipulation Language (DML)** operations such as **INSERT, SELECT, UPDATE, and DELETE** on a database table.

Theory

Data Manipulation Language (DML) is a subset of SQL used to manipulate the data stored in database tables. DML commands allow users to insert new records, retrieve existing records, modify data, and delete records from a table.

The most commonly used DML commands are:

- **INSERT** – Adds new records into a table.
- **SELECT** – Retrieves records from a table.
- **UPDATE** – Modifies existing records.
- **DELETE** – Removes records from a table.

Unlike DDL commands, DML operations affect the data stored in tables and can be rolled back before a COMMIT operation (depending on the DBMS and transaction settings).

Learning Objectives

After completing this experiment, students will be able to:

- Understand DML commands.
- Insert records into a table.
- Retrieve records using the SELECT statement.
- Update existing records.
- Delete records from a table.
- Apply conditional operations using the WHERE clause.

Software Requirements

- Oracle Database
- Oracle SQL*Plus / Oracle SQL Developer

Algorithm / Procedure

1. Open Oracle SQL Developer or SQL*Plus.
2. Connect to the Oracle database.

3. Create the Employee table.
4. Insert sample records.
5. Display all records.
6. Update existing records.
7. Delete specific records.
8. Display the updated table.
9. Stop.

Program

Step 1 : Create Employee Table

```
CREATE TABLE Employee
(
EmpID NUMBER(5),
EmpName VARCHAR2(30),
Department VARCHAR2(30),
Salary NUMBER(10,2)
);
```

Step 2 : Insert Records

```
INSERT INTO Employee
VALUES(101,'Rahul','IT',35000);

INSERT INTO Employee
VALUES(102,'Priya','HR',42000);

INSERT INTO Employee
VALUES(103,'Aman','Finance',39000);

INSERT INTO Employee
VALUES(104,'Sneha','Marketing',45000);

COMMIT;
```

Step 3 : Display All Records (SELECT)

```
SELECT * FROM Employee;
```

Expected Output

EmpID	EmpName	Department	Salary
101	Rahul	IT	35000
102	Priya	HR	42000

EmpID	EmpName	Department	Salary
103	Aman	Finance	39000
104	Sneha	Marketing	45000

Step 4 : Display Specific Columns

```
SELECT EmpName, Salary
FROM Employee;
```

Expected Output

EmpName	Salary
Rahul	35000
Priya	42000
Aman	39000
Sneha	45000

Step 5 : Retrieve Records Using WHERE Clause

```
SELECT *
FROM Employee
WHERE Department='IT';
```

Expected Output

EmpID	EmpName	Department	Salary
101	Rahul	IT	35000

Step 6 : Update Employee Salary

```
UPDATE Employee

SET Salary=50000

WHERE EmpID=101;

COMMIT;
```

Verify Update

```
SELECT * FROM Employee;
```

Expected Output

EmpID	EmpName	Department	Salary
101	Rahul	IT	50000
102	Priya	HR	42000
103	Aman	Finance	39000
104	Sneha	Marketing	45000

Step 7 : Update Department

```
UPDATE Employee

SET Department='Accounts'

WHERE EmpID=102;

COMMIT;
```

Step 8 : Delete Employee Record

```
DELETE FROM Employee

WHERE EmpID=103;

COMMIT;
```

Verify Delete

```
SELECT * FROM Employee;
```

Expected Output

EmpID	EmpName	Department	Salary
101	Rahul	IT	50000
102	Priya	Accounts	42000
104	Sneha	Marketing	45000

Step 9 : Delete Employees from Marketing Department

```
DELETE FROM Employee

WHERE Department='Marketing';

COMMIT;
```

Step 10 : Display Final Table

```
SELECT * FROM Employee;
```

Expected Output

EmpID	**EmpName**	**Department**	**Salary**
101	Rahul	IT	50000
102	Priya	Accounts	42000

Flowchart

Start
│
▼
Create Table
│
▼
Insert Records
│
▼
Display Records
│
▼
Update Records
│
▼
Delete Records
│
▼
Display Final Table
│
▼
End

Explanation

INSERT

Adds new records into the table.

SELECT

Retrieves data from the table.

UPDATE

Modifies existing records.

DELETE

Removes records from the table.

WHERE Clause

Specifies the condition for selecting, updating, or deleting records.

COMMIT

Saves all changes permanently in the database.

Observation

- Employee records were inserted successfully.
- Data was retrieved using the SELECT statement.
- Existing records were updated correctly.
- Records were deleted using the DELETE command.
- Final table reflected all modifications accurately.

Result

The DML commands **INSERT**, **SELECT**, **UPDATE**, and **DELETE** were successfully executed on the Employee table. Records were added, retrieved, modified, and removed as required, demonstrating the effective use of SQL Data Manipulation Language commands.

Applications

- Employee Management Systems
- Student Information Systems
- Banking Applications
- Hospital Management Systems
- Inventory Management Systems
- Railway Reservation Systems
- Library Management Systems

Advantages

- Easy manipulation of database records.
- Supports efficient data retrieval.
- Enables quick modification of existing data.
- Removes unwanted records efficiently.
- Essential for day-to-day database operations.

Viva Questions

1. What is DML?
2. What is the difference between DDL and DML?

3. What is the purpose of the INSERT statement?
4. What is the purpose of the SELECT statement?
5. What is the purpose of the UPDATE statement?
6. What is the purpose of the DELETE statement?
7. What is the WHERE clause used for?
8. What happens if the WHERE clause is omitted in an UPDATE statement?
9. What happens if the WHERE clause is omitted in a DELETE statement?
10. Why is the COMMIT command used after DML operations?

Lab Assignment

1. Create a **Student** table with suitable columns.
2. Insert five student records into the table.
3. Display all student records.
4. Display only the student name and course columns.
5. Update the city of a specific student.
6. Increase the salary of employees belonging to the IT department by ₹5,000.
7. Delete a student record using the Roll Number.
8. Display students whose marks are greater than 75.
9. Compare INSERT, UPDATE, DELETE, and SELECT commands with suitable examples.
10. Explain the role of the WHERE clause in DML operations.

Precautions

- Always verify the table name before performing DML operations.
- Use the **WHERE** clause carefully in UPDATE and DELETE statements to avoid modifying all records.
- Commit the transaction after successful INSERT, UPDATE, or DELETE operations.
- Verify the output after each operation using the SELECT statement.
- Maintain a backup of important data before executing DELETE operations.

Learning Outcome

After completing this experiment, students will be able to:

- Perform basic DML operations on database tables.

- Insert, retrieve, update, and delete records efficiently.
- Use conditional statements with the WHERE clause.
- Understand the importance of transaction control using COMMIT.
- Apply DML commands in real-world database applications such as banking, inventory management, hospital management, and student information systems.